

Prognostic Factors in Melanoma

Age

Males over 60 years of age have the highest mortality rates from melanoma. Older patients tend to present with thicker primary tumors and a higher proportion of ulcerated melanomas. Even after adjusting for these factors, age remains an independent prognostic factor. It has been suggested that age may act as a surrogate for declining host immune defense mechanisms.

Gender

Numerous studies have shown that women have better survival rates than men, even after controlling for tumor thickness and anatomic site.

Anatomic Site

The location of the primary cutaneous melanoma has been variably associated with survival outcomes. In the AJCC study, melanomas on the trunk, head, and neck were linked to a worse prognosis compared to those on the extremities, although the classification grouped sites broadly into (1) extremities or (2) trunk, head, or neck. Another large study of 5093 patients found that melanomas on the back, thorax, upper neck, and scalp (BANS regions) were associated with significantly lower survival rates compared to those on the lower trunk, thigh, lower leg, feet, lower arms, hands, and face. However, anatomic site has not consistently been confirmed as an independent prognostic factor across studies.

Prognostic Factors in Regional Metastasis

The status of the regional lymph nodes is the most powerful prognostic factor for survival in melanoma. The presence of regional lymph node metastasis is associated with a poorer prognosis. The number of involved lymph nodes, regardless of size, is the most significant risk factor.